

Adaptive Autonomous Resource Allocation for 6G V2X Sidelink

Mostafa Thabet

Dept. of Communication Engineering
Military Technical College
Cairo, Egypt
thabet.m@mtc.edu.eg

Ahmed Bendary 

Dept. of Communication Engineering
Military Technical College
Cairo, Egypt
a.bendary@mtc.edu.eg

Ashraf D. Elbayoumy

Dept. of Communication Engineering
Military Technical College
Cairo, Egypt
adiaa@mtc.edu.eg

Abstract—Vehicle-to-Everything (V2X) is set to transform autonomous vehicles by introducing cooperative real-time systems that improve road safety, efficiency, and transportation intelligence. While the Fifth Generation (5G) V2X has seen advances, traditional Sensing-based Semi-Persistent Scheduling (SPS) still faces challenges with collisions due to decentralized resource allocation in sidelink mode 2. This paper proposes an Adaptive Autonomous Resource Allocation (AARA) scheme that allows user equipment to choose resources from a range of adaptive percentages of the lowest power sensed resources, thereby significantly reducing collisions. Our results show that the proposed AARA scheme outperforms conventional SPS, achieving an increase of up to 6% in Packet Reception Ratio (PRR) and extending the communication range to 114%. These insights are crucial for developing efficient and scalable resource allocation strategies relevant to the emerging 6G V2X.

Index Terms—5G NR sidelink, Vehicle-to-Everything (V2X), resource allocation, sensing-based semi-persistent scheduling (SPS), 6G V2X, autonomous vehicles

I. INTRODUCTION

Vehicular communication has become a vital part of the Intelligent Transportation System (ITS), improving safety, traffic management, and collaborative driving. Vehicle-to-Everything (V2X) technology enables Vehicle-to-Vehicle (V2V), Vehicle-to-Pedestrian (V2P) and Vehicle-to-Infrastructure (V2I) communications [1]. With increasing demands for low-latency, real-time, and ultra-reliable communication, V2X supports advanced applications such as high-speed autonomous driving and sensor data exchange [2].

Various wireless technologies have been proposed to enable V2X communications, with a primary focus on Dedicated Short-Range Communications (DSRC) and Cellular-based V2X (C-V2X), each offering dissimilar architectural and performance characteristics. In particular, Fifth Generation New Radio V2X (5G NR-V2X) has emerged as the leading candidate due to its strong support from standardization bodies, scalable system design, and its ability to meet the exact demands of next-generation automotive systems [3].

C-V2X communication has evolved through multiple phases of the Third Generation Partnership Project (3GPP). Fourth Generation Long-Term Evolution V2X (4G LTE-V2X), 3GPP

Release 14, established two resource allocation modes: Mode 3 (controlled through the base station) and Mode 4 (autonomous) [4]. While LTE-V2X worked well for basic safety messaging, its scalability and flexibility limitations led to further advances [3]. The integration of 5G NR V2X in 3GPP Release 16 brought significant improvements, supporting more complex use cases, including extended sensors, vehicular platooning, remote driving, and advanced driving [5].

Autonomous resource allocation modes in 4G LTE-V2X and 5G NR-V2X, referred to as Mode 4 and Mode 2, respectively, enable decentralized sidelink communication in out-of-coverage scenarios [6]. A fundamental commonality is the dependence on Semi-Persistent Scheduling (SPS) based on sensing, where User Equipment (UE) autonomously senses the channel to select and reserve resources for periodic transmissions. In LTE-V2X Mode 4, the SPS mechanism operates within a fixed resource pool using a relatively static and less adaptive configuration. While being effective for basic safety applications, it struggles under high-density or rapidly changing vehicular conditions due to increased risk of collisions and limited flexibility [7].

In contrast, 5G NR V2X Mode 2 enhances the SPS framework with greater adaptability by incorporating flexible numerology, configurable bandwidth parts, and support for periodic and aperiodic traffic [7]. It also introduces advanced features such as resource reevaluation and preemption, allowing UEs to exchange resource usage information and improve coexistence [2]. These enhancements collectively offer more efficient and reliable autonomous resource allocation, allowing more scalable and robust SPS to support the stringent requirements of the next-generation V2X. In [4], they declared that NR can achieve up to a 20% improvement in the Packet Reception Ratio (PRR) compared to LTE. Despite these enhancements in NR-V2X, there exist possible collisions that occur during the selection phase when two or more UEs simultaneously choose the same subchannels for transmission [8].

In this paper, we focus on the (re)selection phase within the legacy SPS mechanism to develop an enhanced autonomous resource allocation strategy for V2X. Positioned within the broader trajectory of the evolution of V2X, we address the limitations of conventional SPS approaches in handling dy-

namic traffic conditions and dense vehicular environments. We present a selection scheme based on the random selection of a percentage of the minimum sensed-power resources within the sensing window. The principal contributions of this work are as follows.

- We present a simple, yet effective, Adaptive Autonomous Resource Allocation (AARA) scheme for enhanced autonomous resource allocation of periodic safety messages.
- We empirically provide the optimal percentage of candidate resources to achieve the maximum PRR, range, and minimum Inter-Packet Gap (IPG).

The remainder of this paper is structured as follows. Section II reviews the legacy Mode 2. Our proposed AARA scheme and simulation setup are introduced in Section III, with numerical results analyzed in Section IV. Section V provides the conclusion.

II. 5G NR V2X MODE-2 OVERVIEW

A. 5G NR Physical Layer

The number of resources that are available for each UE depends heterogeneously on the physical layer resources. 5G V2X utilizes Orthogonal Frequency Division Multiplexing (OFDM) in the physical layer and Single Carrier Frequency Division Multiple Access (SC-FDMA) in the MAC layer, while sidelink communications specifically adopt Cyclic Prefix-OFDM (CP-OFDM), which offers improved latency, spectral efficiency, and resilience to Doppler shift and phase noise [9]. The system supports various Modulation Coding Schemes (MCS) that enable high data rates [10]. Different OFDM numerologies can be achieved with a scalable SCS (Sub-Carrier Spacing) driven by $2^\mu \times 15$ kHz, where $\mu = 0, 1, 2, 3$ is an SCS configuration factor. A subchannel, which is the smallest unit for data transmission in sidelink communication, consists of 10, 12, 15, 20, 25, 50, 75, or 100 Physical Resource Blocks (PRB). The sidelink radio frame has a duration of 10 ms and comprises 10 subframes of 1 ms. The number of time slots in each subframe depends on the configuration factor μ , where the duration of the time slot is equal to $2^{-\mu}$ ms. Each time slot contains 14 or 12 OFDM symbols when using conventional or extended CP, respectively [2].

B. Resource Allocation in Mode-2

Mode-2 in 5G V2X is activated in the case of out-of-coverage, where each UE autonomously selects resources for data transmission with two options: Dynamic Scheduling (DS) and SPS. DS is favorable for aperiodic traffic in which the UE randomly selects resources just once without reservation [2]. The periodic transmission of messages between vehicles is a key operation for V2X applications, which allows the sharing of vehicle status and information such as position, velocity, and current lane [3]. To ensure the delivery of periodic messages as much as possible, SPS has evolved [6]. SPS consists of three main steps: (1) channel sensing, (2) resource selection, and (3) transmission and resource reservation [4]. When a transmitting UE (TX) starts its resource selection phase at a time instant

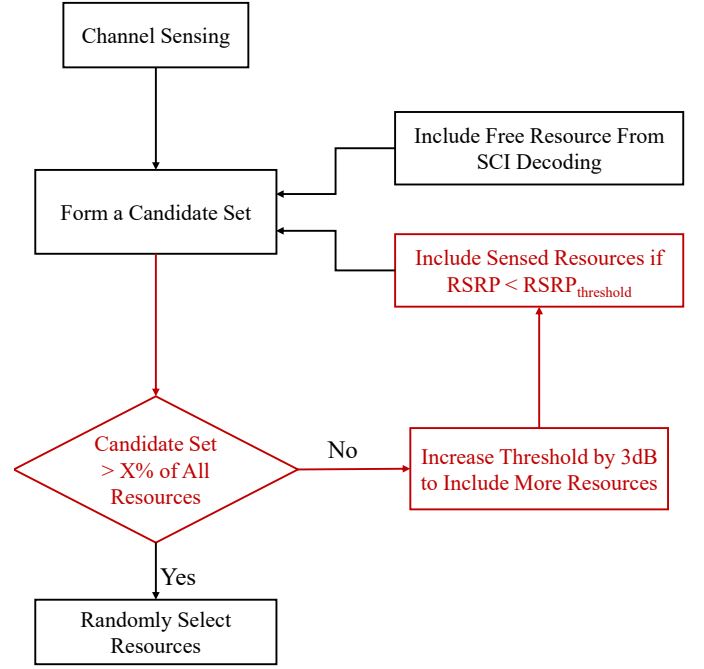


Fig. 1. Resource (re)-selection phase in legacy SPS.

n , it listens to a sensing window before that instance $[n - T_0, n - T_{proc,0}]$ where T_0 is preconfigurable to 1100 ms for periodic transmission or 100 ms for aperiodic transmission and variable $T_{proc,0}$ based on the value of μ . This enables the TX UE to check if the power of each resource in the window is below a specific value called the Reference Signal Received Power (RSRP) threshold, aided by Sidelink Control Information (SCI) decoding to exclude the reserved resources to form a candidate set. If that candidate set contains less than X% of the total resources, the process is repeated after increasing the threshold by 3 dB to include more resources until achieving the condition shown in Fig. 1. The parameter X can be 20, 35, or 50, depending on the packet priority for which the UE executes the resource selection [2].

Next, the UE randomly selects the required resources in the resource selection window $[n + T_1, n + T_2]$, $T_1 \leq T_{Proc,1}$ defined in [5] and depends on μ , T_2 must be included in the range of $[T_{2,min}, PDB]$ where PDB (packet delay budget) is the latency constraint for transmitting a transport block for a corresponding V2X application and $T_{2,min}$ is a predefined factor depending on μ . Before using the selected resource for transmission, the UE undergoes a resource reevaluation and pre-emption process to ensure availability of the preselected resource, avoid possible congestion, and give preference to the high-priority packet [6]. In the last phase, the UE utilizes the selected resources for a periodic interval RRI (Resource Reservation Interval) and for a specific number of times RC (Reselection Counter). UE selects new resources (reselection) whenever the RC is set to zero with probability $1 - P_{keep}$ in addition to the reselection triggered by the pre-emption process.

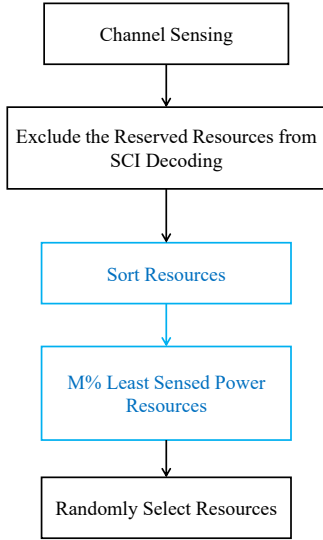


Fig. 2. Proposed AARA resource (re)-selection phase.

III. RESEARCH PROBLEM AND THE PROPOSED SCHEME

The performance evolution of the (re)selection phase within the SPS mechanism from 4G LTE to 5G NR has been the focus of extensive research in previous work [4, 7]. In 5G NR, improvements such as resource reevaluation and blind retransmission have been introduced to mitigate persistent reliability challenges in distributed resource allocation [5]. Nevertheless, the problem of collisions remains unresolved, primarily due to the stochastic nature of the resource reselection process following the sensing phase. Since UEs autonomously and independently execute random resource choices, the likelihood that multiple interfering UEs simultaneously select identical resources becomes significant, leading to an elevated probability of collision [8, 11, 12]. This intrinsic gap of SPS highlights the limitations of random resource selection and underscores the need for more coordinated allocation strategies to meet stringent reliability and latency requirements in V2X communications.

We propose the AARA scheme, a simple yet effective resource (re)selection mechanism designed to address two major limitations of legacy SPS: collisions resulting from threshold-based sorting and delay due to repeated formation of the candidate set caused by an insufficient number of candidate resources. The general structure of the proposed scheme is illustrated in Fig. 2, and its operation is described as follows. After the initial sensing phase, each UE sorts candidate resources, excluding the reserved ones, through SCI decoding. Due to the unique positional attributes of individual vehicles, the power measurements within the sensing window naturally differ between UEs. To exploit this diversity, each UE randomly selects the required resources from the lowest $M\%$ of the resources based on the measured power levels within the sensing window. By restricting the selection to minimally powered resources, the proposed AARA approach achieves a

TABLE I
SIMULATION PARAMETERS

Scenario	
Road layout	Highway, 3+3 lanes
Vehicle Density (ρ)	50,150 vehicles/km
Average speed	70 km/h
Speed standard deviation	7 km/h
Propagation Model	
Frequency band	operate at 5.9 GHz
Bandwidth	10 MHz
Noise figure	6 dB
Transmission power	23 dBm
Antenna gain (tx and rx)	3 dBi
Propagation model	WINNER+, Scenario B1
Shadowing	Var. 3 dB, decorr. dist. 25 m
Data Traffic	
Packet generation period	100 ms
Packet size	190,350,1000 bytes
Access Layer (NR-V2X)	
MCS	11 (16QAM, $R_c \approx 0.3$)
Subchannel size	10 PRB
SCS	15 kHz
RSRP threshold	-126 dBm
Persistence probability	0.4
Min. time for the allocation, T_1	1 ms
Max. time for the allocation, T_2	100 ms
Max. re-evaluation time, T_3	3 ms
Monte Carlo Simulation	
Simulation time	300 sec
Number of iterations	15

more efficient use of the available spectrum and effectively reduces the likelihood of collision.

Note that the sorting operation requires more processing than the threshold-based technique. However, the vehicle's nature overcomes this requirement by providing higher processing and memory capabilities. The percentage M is not fixed and may be used differently to classify priority levels. Rather, an optimal value of M will be determined globally as in Section IV.

IV. SIMULATION SETUP AND PERFORMANCE EVALUATION

The numerical results in this paper are obtained using the WiLabV2Xsim simulator [10]. The simulation parameters are described in Table I, adopted from [13], and are further detailed below for clarity. The simulation models a bidirectional highway comprising three lanes per direction. Two vehicle densities are evaluated, 50 and 150 vehicles/km, and each vehicle travels at a speed modeled by a Gaussian distribution, with a mean of 70 km/h and a standard deviation of 7 km/h. Communication occurs over the 5.9 GHz ITS band with a 10 MHz bandwidth. Transmit power is set to 23 dBm, with 3 dBi antenna gains at both ends and a 6 dB receiver noise figure. The WINNER+ scenario B1 model is applied [10], including log-normal shadowing of 3 dB and a decorrelation distance of 25 m.

Vehicles generate periodic broadcast packets every 100 ms. Packet sizes of 190, 350, and 1000 bytes are used, corresponding to standard vehicular messages such as the Basic Safety Message (BSM) and the Cooperative Awareness Message (CAM) [3, 6]. The sidelink operates with a 15 kHz

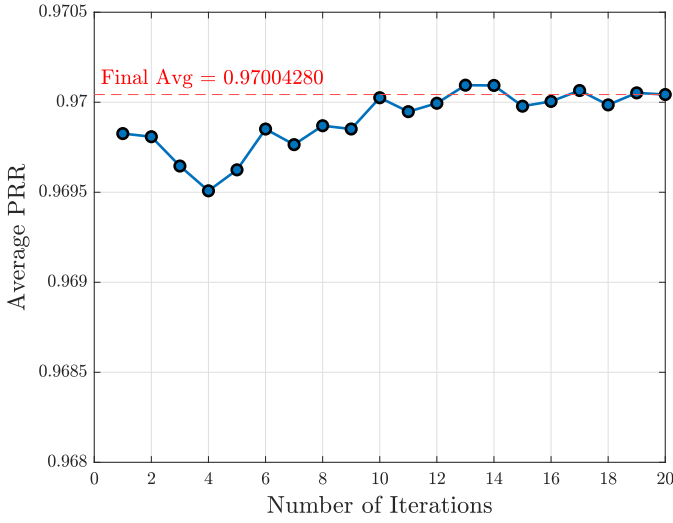


Fig. 3. Convergence analysis shows the validation of the selection of the number of iterations required for robust simulation results.

sub-carrier spacing and MCS index 11 (16QAM, coding rate $R_c \approx 0.3$). Each subchannel consists of 10 PRBs. The sensing-based SPS mechanism is applied with an RSRP threshold of -126 dBm. The allocation timers are set to $T_1 = 1$ ms, $T_2 = 100$ ms, and $T_3 = 3$ ms. To determine the minimum number of iterations required for statistically reliable results in a 300-second simulation, a convergence analysis was performed using average PRR as the performance metric for legacy SPS in the scenario of a 350-byte message and a vehicle density of 50 vehicles/km. As shown in Fig. 3, the average PRR exhibits a convergence behavior that begins at approximately 10 iterations. To enhance confidence in the simulation results, a conservative choice of 15 iterations was adopted for all experiments.

Performance evaluation is based on the following metrics [14]. (1) **PRR** is the fraction of UEs that successfully receive packets among all UEs located at that specific distance from the transmitter. For the calculation of the average value, we consider the range of awareness (RAW) as 150 meters, representing the distance within which a PRR target must be maintained (2) **IPG** is defined as the duration, measured in milliseconds, between two consecutive data packets that are successfully received by a particular receiver from a specific transmitting source in a particular range. IPG indicates the reliability of packet reception and data freshness, which is related to the Age of Information (AoI) at the receiver. (3) **Communication range** represents the greatest distance at which the PRR exceeds a specified threshold (0.95 in this case). This reflects the longest distance for which packets can be received with reliable delivery performance.

A. Numerical Results

With the legacy 5G V2X SPS as our benchmark, in this section, we introduce comprehensive numerical results to evaluate the performance and effectiveness of our proposed AARA scheme in various scenarios. Many parameters affect

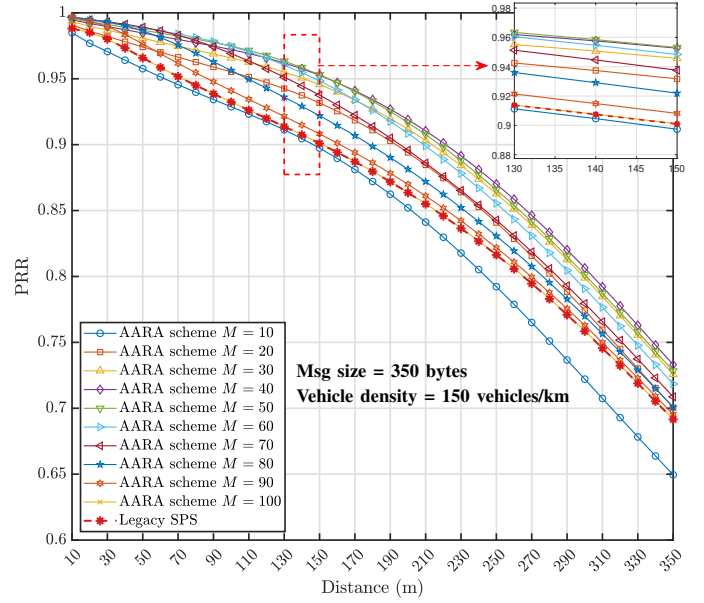


Fig. 4. Our proposed AARA scheme at optimum percentage $M = 40\%$ is superior over legacy SPS for all distances.

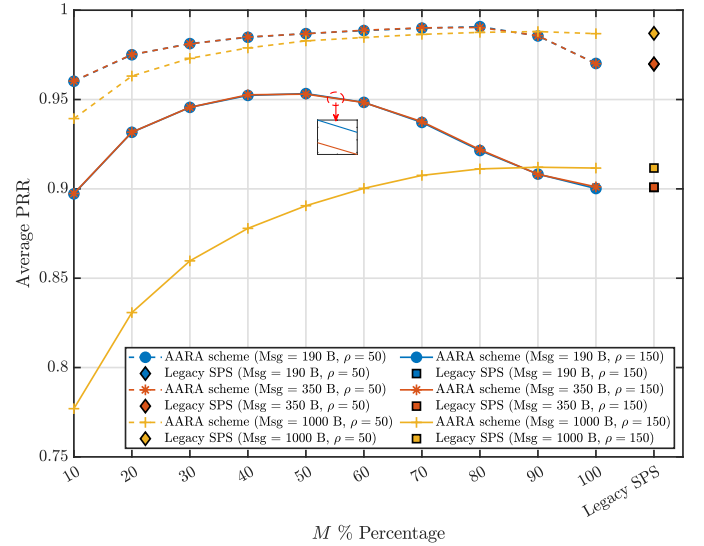


Fig. 5. Our proposed AARA scheme predominates the legacy SPS, achieving higher PRR at optimum percentage M for all scenarios.

the performance of resource allocation. The selection phase depends directly on the density of the vehicle. The size of a message directly influences how many resources are required for its transmission. Given that bandwidth and MCS affect the number of resources available, both parameters are fixed in this work to ensure fairness between evaluations.

In the WiLabV2Xsim simulator [10], PRR is plotted, representing the PRR value for a distance difference. For example, when evaluated at 70 m, PRR reflects the ratio of successfully received packets to total transmissions within the distance window of 60 to 70 m. Unlike these procedures to validate

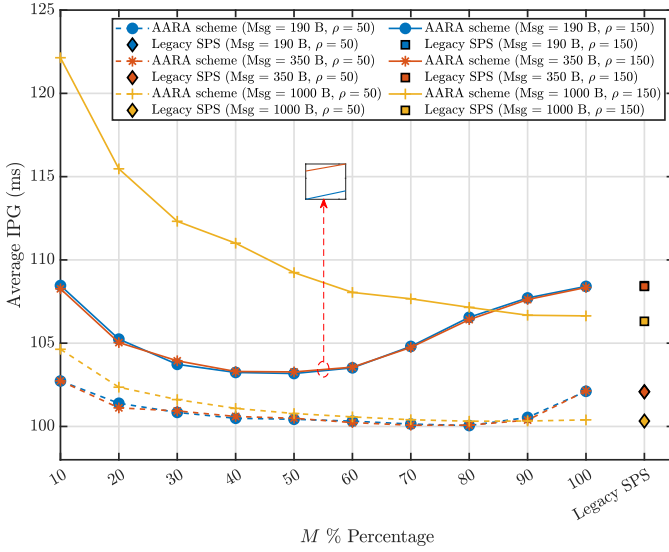


Fig. 6. Minimum IPG is achieved at optimum M percentage for all scenarios.

the performance of our proposed AARA scheme, we plot the cumulative PRR versus distance, which represents the average PRR value in different RAW. Fig. 4 shows the case of a 350-byte message size at a high density of 150 vehicles/km, illustrating how our proposed AARA scheme at optimal $M = 40\%$ performs consistently across increasing distances, while legacy SPS performance achieves low PRR.

It can be seen that for message sizes of 190 and 350 bytes, the performance appears to be the same due to the granularity constraint, since the smallest resource allocation unit is a subchannel, and the restriction in transport block size determination the number of subchannels required for packet transmission of both message sizes is the same according to the calculations in [2] and the procedures described by 3GPP in [15].

As shown in Fig. 5, for 190- and 350-byte messages, a clear superiority of our proposed AARA scheme over the legacy SPS is evident in both cases of vehicle density. The average PRR as a function of the percentage M shows a distinctive concave pattern, reaching its optimum at $M = 80\%$ for the 50 vehicle/km case with about 3% higher PRR than legacy SPS and at $M = 40\%$ for the 150 vehicles/km case with an enhancement of about 6%. This optimal value is also reflected in the corresponding IPG performance in Fig. 6, which exhibits a complementary convex pattern, achieving its minimum delay at the same optimal M percentage with minimizing IPG by about 3% for 50 vehicles/km case and 5% for 150 vehicles/km case compared to legacy SPS, reaffirming the direct correlation between PRR maximization and IPG minimization [14].

For the case of larger packets (1000 bytes) in both density scenarios, the number of subchannels needed for packet transmission is greater than the 190 and 350-byte message cases, which leads to a lower amount of usable broadcast resources within the selection window [2]. The performance of the proposed AARA scheme is closely similar to that of the legacy

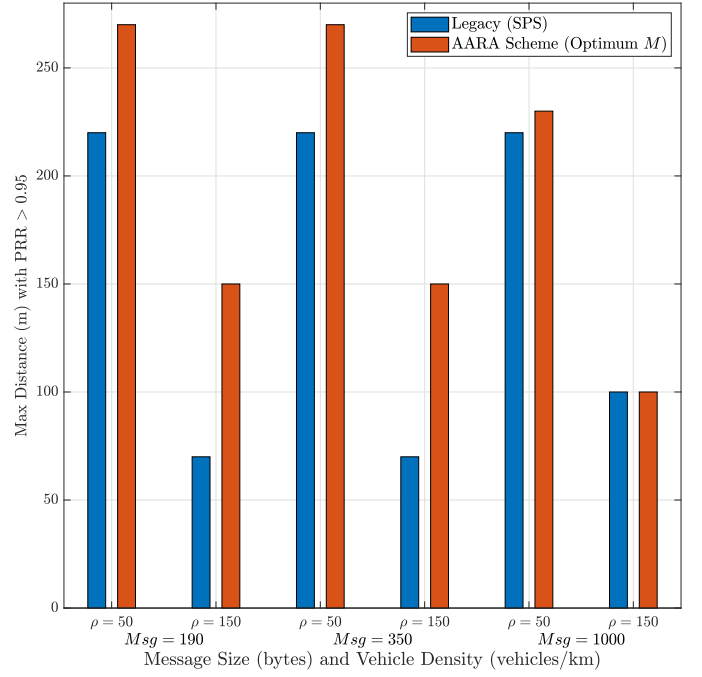


Fig. 7. Our proposed AARA scheme enlarges the maximum distance in all the scenarios, providing a higher range than legacy SPS with $PRR \geq 0.95$.

SPS at $M = 90\%$. This reveals that for large message sizes, the optimal candidate set is all available resources without sorting or employing a threshold. These observations suggest the existence of an optimal percentage, dependent on scenario, vehicle density, and message size, to achieve the best overall performance under varying conditions.

Fig. 7 illustrates the maximum communication distance achieved with a PRR threshold of 95%. To benchmark the improvements, we contrast the performance of legacy SPS with our proposed AARA approach at optimal M percentages. The maximum reliable communication range decreases with increasing vehicle density due to increased channel contention and interference. Specifically, the AARA scheme achieves notable gains compared to the legacy baseline, increasing the reliable communication range by about 23% in low-density scenarios and by 114% in high-density scenarios.

V. CONCLUSION AND FUTURE WORK

Our proposed AARA scheme is a simple yet effective resource (re)-selection scheme that aims to improve autonomous resource allocation for future sidelink mode-2. The approach departs from threshold-based selection loops by introducing a lowest-power selection strategy, whereby UEs probabilistically select resources from among those exhibiting the lowest sensed power levels. We introduced a cumulative PRR metric that offers a more detailed and informative evaluation of performance compared to traditional metrics. Through extensive simulation in highway scenarios, with varying vehicle densities and message sizes, our proposed AARA scheme demonstrated consistent gains over conventional legacy SPS in terms of PRR, IPG, and range, indicating that for large message sizes,

the optimal strategy is to use all available resources without applying sorting or thresholds. Furthermore, the results revealed a clear dependency between network load, message size, and optimal selection percentage M , confirming the relevance of adaptive strategies in resource-constrained V2X environments.

The results provide valuable guidance for developing decentralized adaptive resource allocation strategies in future 6G V2X networks, supporting the goals of ultra-reliability, low-latency, and intelligent operation. Based on this foundation, future work will establish a comprehensive mathematical model that includes PHY / MAC parameters, vehicular dynamics, and traffic variability and will take advantage of learning-based techniques, such as reinforcement learning, to enable real-time and adaptive resource allocation.

REFERENCES

- [1] A. Alalewi, I. Dayoub, and S. Cherkaoui, "On 5G-V2X use cases and enabling technologies: A comprehensive survey," *IEEE Access*, vol. 9, pp. 107710–107737, 2021.
- [2] M. H. C. Garcia, A. Molina-Galan, M. Boban, J. Gozalez, B. Coll-Perales, T. Sahin, and A. Kousaridas, "A tutorial on 5G NR V2X communications," *IEEE Communications Surveys and Tutorials*, vol. 23, no. 3, pp. 1972–2026, 2021.
- [3] J. Clancy, D. Mullins, B. Deegan, J. Horgan, E. Ward, C. Eising, P. Denny, E. Jones, and M. Glavin, "Wireless access for V2X communications: Research, challenges and opportunities," *IEEE Communications Surveys and Tutorials*, vol. 26, no. 3, pp. 2082–2119, 2024.
- [4] C. Shin, E. Farag, H. Ryu, M. Zhou, and Y. Kim, "Vehicle-to-everything (V2X) evolution from 4G to 5G in 3GPP: Focusing on resource allocation aspects," *IEEE Access*, vol. 11, pp. 18689–18703, 2023.
- [5] "Study on NR Vehicle-to-Everything (V2X) TR 38.885 (Release 16)," 2019.
- [6] TSGR, "TR 137 985 - V17.1.1 - LTE; 5G; overall description of radio access network (RAN) aspects for vehicle-to-everything (V2X) based on LTE and NR (3GPP TR 37.985 version 17.1.1 release 17)," 2022.
- [7] K. Sehla, T. M. T. Nguyen, G. Pujolle, and P. B. Velloso, "Resource allocation modes in C-V2X: From LTE-V2X to 5G-V2X," *IEEE Internet of Things Journal*, vol. 9, no. 11, pp. 8291–8314, 2022.
- [8] C. Brady and S. Roy, "Opportunistic retransmissions: A novel distributed adaptive algorithm for NR C-V2X mode 2," *IEEE Systems Journal*, vol. 18, no. 2, pp. 881–892, 2024.
- [9] M. Tabassum, F. H. Bastos, A. Oliveira, and A. Klautau, "NR sidelink performance evaluation for enhanced 5G-V2X services," *Vehicles*, vol. 5, no. 4, pp. 1692–1706, 2023.
- [10] V. Todisco, S. Bartoletti, C. Campolo, A. Molinaro, A. O. Berthet, and A. Bazzi, "Performance analysis of sidelink 5G-V2X mode 2 through an open-source simulator," *IEEE Access*, vol. 9, pp. 145648–145661, 2021.
- [11] V. Todisco, C. Campolo, A. Molinaro, A. O. Berthet, R. A. Stirling-Gallacher, and A. Bazzi, "Full duplex based collision detection to enhance the V2X sidelink autonomous mode," *Computer Networks*, vol. 254, 2024.
- [12] Annu and P. Rajalakshmi, "Towards 6G V2X sidelink: Survey of resource allocation—mathematical formulations, challenges, and proposed solutions," *IEEE Open Journal of Vehicular Technology*, vol. 5, pp. 344–383, 2024.
- [13] 5GAA, "White paper: 5G V2X direct communication evaluation approach 2," 2024. Accessed: Jul. 2024. [Online]. Available: <https://5gaa.org/content/uploads/2024/07/5gaa-wi-nr-v2x-eval.pdf>.
- [14] J. Yin and S. H. Hwang, "Reuse distance-aided resource selection mechanisms for NR-V2X sidelink communication," *Sensors*, vol. 24, no. 1, 2024.
- [15] 3rd Generation Partnership Project (3GPP), "TS 138 214 - V16.2.0 - 5G; NR; physical layer procedures for data (3GPP TS 38.214 version 16.2.0 release 16)," tech. rep., ETSI, 2020.